Measurement CHEAT SHEET

UNITS AND THE SI

7 base units: m · kg · s · A · K · mol · cd (SI)

derived: m/s, m^2, N = kg·m/s^2, J = N·m, W = J/s

prefixes: M 10^6 · k 10^3 · m 10^{-3} · μ 10^{-6} · n 10^{-9}

105 m = 1.05×10^2 m; 0.0001 m = 1×10^{-4} m

convert: multiply by one, $72 \text{ km/h} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = 20 \text{ m/s}$

squared units, squared factor: 1 m^2 = 10,000 cm^2

ROUNDING AND CALCULATIONS

round half up: dropped digit ≥ 5 raises the last kept

2.746 → 2.75 · 2.744 → 2.74 · 1.996 → 2.00 (3 s.f.)

exact five: 2.745 → 2.75 (half up) · 2.74 (half to even)

round once: 2.4449 → 2.4 (in steps: 2.5, wrong)

× ÷ keep the fewest figures: 100.0/11.27 = 8.873

+ − keep the fewest decimals: 12.6 + 0.247 + 3.15 = 16.0

close numbers: 10.48 − 10.47 = 0.01 (one figure left)

SIGNIFICANT FIGURES

the last digit you write is the first you're unsure of

least count: one step of the instrument = ± one reading

1 non-zero digits count: 14.32 → 4

2 zeros between count: 2.05 → 3, 1005 → 4

3 leading zeros never: 0.0045 → 2

4 trailing zeros after a point count: 2.50 → 3

5 a whole number's trailing zeros don't: 4500 → 2;
  4.50×10^3 → 3

exact numbers (counts, the 2 in 2πr) never limit

new unit, same count: 14.32 cm = 0.1432 m → 4

EDGE TESTS

B exact: ΔZ = ΔA; close values: 0.01 ± 0.02 s, not real

÷ exact 20: 35.9 ± 0.2 s → 1.795 ± 0.01 s (same %)

√ halves the %, 1/A keeps it: √(900 ± 6) = 30.0 ± 0.1

UNCERTAINTY IN A RESULT

absolute ΔA · relative ΔA/A · percentage 100 ΔA/A

14.3 ± 0.1 cm: 0.7 %; repeated: mean ± half the range

  0.62 0.65 0.65 0.66 0.62 s → 0.64 ± 0.02 s

Z = A ± B: ΔZ = ΔA + ΔB (absolute uncertainties add)

  14.3 ± 0.1 − 8.6 ± 0.1 = 5.7 ± 0.2

Z = AB or A/B: $\frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$ (relative add)

  14.3 × 7.0 (± 0.1 each) = 100 ± 2 cm^2

$Z = A^n$: $\frac{\Delta Z}{Z} = n \frac{\Delta A}{A}$; tile 30.0 ± 0.1 → 900 ± 6

quote: Δ to 1 s.f., the value to the same place

random errors scatter (precision); systematic shift (accuracy)

Measurement CHEAT SHEET

DIMENSIONS

$[\text{speed}] = \mathrm{L\,T^{-1}}$ · $[\text{acceleration}] = \mathrm{L\,T^{-2}}$

$[\text{force}] = \mathrm{M\,L\,T^{-2}}$ · $[\text{work, energy}] = \mathrm{M\,L^2\,T^{-2}}$

$[\text{power}] = \mathrm{M\,L^2\,T^{-3}}$ · $[\text{pressure}] = \mathrm{M\,L^{-1}\,T^{-2}}$

$[\text{density}] = \mathrm{M\,L^{-3}}$ · $[G] = \mathrm{M^{-1}\,L^3\,T^{-2}}$

dimensionless: angles (L/L), ½, 2π, ratios

units say how big; dimensions say what kind

BEFORE YOU TRUST IT

before you trust a result:

1 does every number carry its unit?

2 does it keep only the digits it has earned?

3 is the difference bigger than its uncertainty?

4 could a systematic error hide outside the ±?

DIMENSIONAL ANALYSIS

check: the same dimensions on both sides, every term

$v^2 = u^2 + 2as$ passes · $s = ut + at$ fails · $s = ut + at^2$ passes, but the true one has ½: dimensions miss pure numbers

convert: $n_2 = n_1\,(M_1/M_2)^x\,(L_1/L_2)^y\,(T_1/T_2)^z$

1 N = 10^5 g·cm/s²; 1 lbf = 0.45359237 × 9.80665 = 4.448 N

find: $T = k\,m^x\,\ell^y\,g^z \to x = 0,\ y = ½,\ z = -½$: $T = k\sqrt{(\ell/g)}$

simulated 1.00 m: 2.007 s → $k = 2\pi$; 60° swing: +7 %

$g = 4\pi^2\ell/T^2$: 0.800 m, T = 1.795 ± 0.01 s → 9.8 ± 0.1 m/s²

IN PYTHON

round(x, n − 1 − floor(log10(abs(x))))

rounds x to n significant figures: 0.004567 → 0.0046

mean(xs) · half range (max(xs) − min(xs))/2

Decimal("2.745").quantize(Decimal("0.01"), ROUND_HALF_UP) → 2.75; round(2.675, 2) gives 2.67 (binary!)